

RAKSHITHA UMMADISETTI

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SUMMARY

Generative AI Engineer with **1.5+ years of experience designing, developing, and deploying production-grade AI systems**. Specialized in agentic AI architectures, multi-agent orchestration (LangGraph), Retrieval-Augmented Generation (RAG) pipelines, Large Language Model (LLM) integration, Model Context Protocol (MCP) tool development, and embedded machine learning on edge hardware. Proven track record delivering end-to-end AI products spanning enterprise automation, hardware-integrated intelligence, and biometric inference systems. Proficient in **Python, FastAPI, Docker, LangChain, OpenAI GPT-4, LangGraph**, and cloud-integrated MLOps pipelines.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript (React.js)
Generative AI	Large Language Models (LLMs), RAG, LangChain, LangGraph, Prompt Engineering, Structured Output Generation, LLM Fine-Tuning, OpenAI GPT-4, Claude API, Llama
Agentic AI	Multi-Agent Orchestration, Model Context Protocol (MCP), Tool Calling / Function Calling, Agent Planning & Reflection, Self-Critique Loops, Autonomous Workflow Design
Retrieval & Search	Vector Databases (Pinecone, FAISS, Chroma), Hybrid Search (BM25 + Dense Retrieval), Semantic Search, Embeddings, Cross-Encoder Re-ranking
ML / AI	NLP, Computer Vision, Graph Attention Networks (GAT), Embedded AI / Edge ML, Biometric Signal Processing (HRV, SpO2, EDA), TensorFlow, PyTorch, Scikit-learn, ONNX, TFLite
Infrastructure	FastAPI, Docker, Git, GitHub Actions, REST APIs, MLOps, CI/CD Pipelines, Firebase Realtime Database

EXPERIENCE

Generative AI Software Development Engineer Tropoleap	Oct 2025 – Present <i>Hyderabad, India</i>
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- Architected a **production Model Context Protocol (MCP) server** integrated with Rally, exposing 12+ standardized tools enabling LLM agents to autonomously create, query, update, and reprioritize work items — **eliminating ~8 hrs/week** of manual ticket grooming per team.
- Designed a **multi-agent orchestration framework** (planner, retriever, executor, validator) using **LangGraph** that decomposes complex business workflows into cooperating specialist agents; **increased task-completion accuracy by 35%** and **reduced human-in-the-loop intervention by 50%**.
- Built **hybrid RAG pipelines** (dense vector search + BM25 + cross-encoder re-ranking) grounding LLM outputs in proprietary knowledge bases; **reduced hallucination rate by ~40%** versus a naive embedding-only baseline.
- Integrated learned ML models with a deterministic rule engine on a rapid soil-testing device (7-in-1 NPK + NIR spectral sensors), delivering nutrient predictions with **<5% RMSE** on field validation data.
- Built end-to-end **biometric ML pipelines** for a wearable mental health platform, processing HRV, SpO2, and EDA signals to classify stress states with **87%+ accuracy**, surfacing early-warning patterns for clinical teams.

AI/ML Engineer Intern iBridge LLC	Feb 2025 – Sep 2025 <i>Remote</i>
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- Built a **contract clause-extraction pipeline** using RAG and specialized LLM agents (extractor, classifier, summarizer), **reducing manual legal review time by ~70%** across 500+ contracts.
- Automated invoice data extraction using a **Donut transformer model** (OCR-free document understanding), **cutting manual data-entry effort by 85%** across thousands of monthly invoices.
- Shipped a **hybrid recommendation engine** (embeddings + collaborative filtering) and an NLP email-classification pipeline automating routing and prioritization of **10,000+ emails/month**.

Machine Learning Engineer Intern

Jul 2024 – Dec 2024

Eizen.ai

Hyderabad, India

- **Improved autism-detection model accuracy by 30%** through custom feature engineering and model ensembling; deployed for real-time inference with **sub-100ms latency using ONNX**.
- Built a **multilingual dubbing platform** (speaker diarization → translation → voice cloning → lip-sync) supporting 6 Indian languages, reducing dubbing turnaround **from days to hours**.
- Trained domain-specific **private LLMs** and applied **Graph Attention Networks (GAT)** to construct relational event graphs, surfacing non-obvious entity relationships for client decision-making dashboards.

PROJECTS

- **EB-1/EB-2 Visa Eligibility Advisor.** Multi-agent system (criteria-matcher, evidence-gap analyzer, strategy-planner) that ingests a candidate's resume and portfolio, evaluates alignment with USCIS EB-1A/EB-2 NIW criteria, and outputs a structured eligibility report with a prioritized 6-month evidence-building roadmap. FastAPI backend, React.js dashboard; **reduced self-assessment time from ~3 hrs to under 10 minutes** in user testing.
- **AgentEval — Open-Source Multi-Agent Evaluation Harness.** Benchmarking framework measuring agentic pipeline quality across task-completion accuracy, hallucination rate, and tool-call efficiency. Ships with 50+ canonical test cases spanning RAG, function calling, and self-critique loops; supports LangGraph and custom orchestrators. **Reduced debugging turnaround by 60%** versus ad-hoc evaluation in A/B tests.
- **Multi-Agent Research Assistant.** Multi-agent pipeline (planner, retriever, synthesizer) for complex multi-hop QA over custom document corpora. Hybrid BM25 + dense retrieval with cross-encoder re-ranking; **improved grounded-answer accuracy by 28%** over a single-agent RAG baseline.
- **StockChat.** Real-time inventory intelligence system detecting understock, overstock, and stock-out conditions across distributed warehouses. LangChain + OpenAI GPT-4 backend with a natural-language query interface — no SQL required for non-technical ops teams.
- **Intent Email.** NLP-powered email-management app classifying intent and converting emails into structured actionable tasks. React.js front end, FastAPI backend, SQL storage; handles 15+ intent categories.

EDUCATION

Bachelor of Technology, Artificial Intelligence & Data Science

Dec 2021 – May 2025

B V Raju Institute of Technology (CGPA: **9.02 / 10**)

Telangana, India

Relevant Coursework: Deep Learning, NLP, Computer Vision, Data Structures & Algorithms

AWARDS, ACHIEVEMENTS & PUBLICATIONS

- **Runner-up**, Graphethon 2024 — 72-Hour National Hackathon (500+ participants), India
- **1st Runner-Up**, Code for Good Hackathon — HST 2023, India (tech solutions for nonprofits)
- **Finalist**, Wall Street Analytics Challenge (ATMOS-2023) — complex financial analytics under time pressure
- **Finalist**, DST Geospatial Hackathon 2023 (Government of India sponsored), India
- **“Quantum-Inspired ML Models for Cyber Threat Intelligence”** — ICICBDA-2024 (peer-reviewed)
- **“Classification of Brain MRI Images Using ROI-Based CNN”** — International Conference on Intelligent Healthcare and Computational Neural Modelling (peer-reviewed)